Modified Duke Criteria — Infective Endocarditis

suspected IE · adults and children (limitations in children)

Duke IE

MAJOR CRITERIA ✚

- **Positive blood culture:** typical organisms in 2 separate cultures (S. viridans incl. nutritional variants, S. bovis, HACEK, S. aureus, or community enterococci without a primary focus); OR persistent bacteraemia (cultures >12 h apart, or all 3 / majority of ≥4 with first and last ≥1 h apart); OR Q fever serology / single positive culture for Coxiella burnetii or anti-phase-1 IgG >1:800 ✚
- **Endocardial involvement:** positive echo (oscillating intracardiac mass on valve, supporting structures, in regurgitant jet path or on implanted material; abscess; new partial dehiscence of a prosthetic valve) OR new valvular regurgitation ✚

MINOR CRITERIA ✚

- Predisposing heart condition or IV drug use ✚
- Fever ≥38 °C ✚
- Vascular: major arterial emboli, septic pulmonary infarcts, mycotic aneurysm, intracranial haemorrhage, conjunctival haemorrhages, Janeway lesions ✚
- Immunologic: Osler's nodes, Roth spots, glomerulonephritis, rheumatoid factor ✚
- Microbiologic: positive culture not meeting a major criterion, or serology of active infection with an IE organism ✚

DIAGNOSIS ✚

Definite IE Pathology (organisms in vegetation/emboli/abscess, or histology of active IE) OR 2 major OR 1 major + 3 minor OR 5 minor ✚

Possible IE 1 major + 1 minor OR 3 minor ✚

Rejected Firm alternative diagnosis; resolution with ≤4 days of antibiotics; or no pathological IE at surgery/autopsy after ≤4 days of antibiotics ✚

SA STG — WORKUP

3 blood cultures from different sites within 2 h if very ill, otherwise within 24 h; little gain from >5 ✚

Suspect IE in every child with fever and underlying heart disease ✚

Children: criteria rely on rare features; splenomegaly, clubbing, haematuria, CRP/ESR are not included ✚

NOTE

- 2023 Duke-ISCVID update (not in STG): adds cardiac CT / PET-CT and intra-operative findings, expands typical organisms

PAED STG & EML Paediatric Hospital Level, 2025-2029 cycle (updated Mar 2026) — pp 4.10, 4.11, 4.9 · **ISCVID** Fowler VG et al. 2023 Duke-ISCVID criteria. Clin Infect Dis 2023;77:518-26